

STATE PATTERN

Behavioural Design Pattern

State Pattern

Imagine you're creating a **DirectionService** class for a maps application like Google Maps. This service needs to calculate **how long (ETA)** it will take to reach your destination and what **route (directions)** to follow between two locations. The travel time and route change depending on your chosen mode of transport:

- Walking
- Cycling
- Car
- Train

Problems in our Code

Messy if-else Chains and Tight Dependencies:

- The *DirectionService* class relies heavily on **if-else** statements to check which transportation mode is active and then calculate travel time and routes accordingly.
- When you keep adding more transportation options, these conditional checks grow longer and more complicated, making the code difficult to read and update.

Breaking the Open/Closed Principle:

- Introducing a new transportation mode (like Airplane or Boat) means you have to go back and modify the *DirectionService class* itself, which violates the **Open/Closed Principle** (code should be open for adding features but closed for changing existing code).

State Pattern: Structure

Context: This is the main class (like *DirectionService*) that keeps track of which state is currently active.

State: This is the interface that defines what methods all states must have (like calculating ETA or getting directions).

Concrete State: These are the actual state classes (like WalkingMode, CarMode, TrainMode) that implement the State interface. Each one represents a different mode or condition of the Context object.

Examples

Music Player: A music player app that behaves differently depending on its current state (**playing**, **paused**, or **stopped**). The play button does different things in each state.

Order Tracking System: An e-commerce order that moves through different states like Order Placed, Packing, Shipped, Out for Delivery, and Delivered. Each state has different actions available.

ATM Machine: An ATM that changes behavior based on its state (idle, card inserted, PIN verification, selecting transaction, dispensing cash). Each state allows different operations.

Traffic Light System: A traffic signal that cycles through states (red, yellow, green) where each state determines what vehicles can do and automatically transitions to the next state after a timer.